Lab 5 Differential Expression and Functional Annotation

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Gene Expression and Data Visualization

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Lab 5:

In this lab, we will be conducting a two-sample test for each gene/probe on the array to identify differentially expressed genes/probes between ketogenic rats and control diet rats. This small data set was run on the rat RAE230A Affymetrix array. The objective of the study was to determine differences in mRNA levels between brain hippocampi of animals fed a ketogenic diet (KD) and animals fed a control diet. “KD is an anticonvulsant treatment used to manage medically intractable epilepsies”, so differences between the 2 groups of rats can provide biological insight into the genes that are regulated due to the treatment.

We are going to identify those genes/probes that are differentially expressed between the 2 rat diet groups and plot the results with a couple of different visual summaries.

Q1) Download the GEO rat ketogenic brain data set and save as a text file.

Q2) Load into R, using read.table() function and header=T/row.names=1 arguments.

▼ Code

```
# Load the dataset into R
dat <- read.table("rat_KD.txt", header = TRUE, row.names = 1, sep = "\t", check.names = FALSE)

# Check dimensions and first few columns
dim(dat)
```

```
[1] 15923    11
```

▼ Code

```
head(dat[, 1:5])
```

	control.diet.19300	control.diet.19301	control.diet.19302
AFFX-BioB-5_at	76.4424	86.0648	80.6466
AFFX-BioB-M_at	94.7098	73.4476	88.6791
AFFX-BioB-3_at	27.8835	44.5481	33.8824
AFFX-BioC-5_at	174.3390	151.8240	167.4290
AFFX-BioC-3_at	87.0285	94.0345	120.2830
AFFX-BioDn-5_at	150.1180	135.4670	163.3840

	control.diet.19303	control.diet.19304
AFFX-BioB-5_at	93.8439	73.1219
AFFX-BioB-M_at	111.5530	92.1317
AFFX-BioB-3_at	60.0250	39.2463
AFFX-BioC-5_at	200.4780	170.7280
AFFX-BioC-3_at	114.5800	100.1920
AFFX-BioDn-5_at	170.9000	155.7970

Q3) First log2 the data, then use the Student's t-test function in the notes to calculate the changing genes between the control diet and ketogenic diet classes. (Hint: use the names() function to determine where one class ends and the other begins).

▼ Code

```
## Step 1: Log2-transform the expression values (robust)
# Coerce to numeric matrix and avoid log2(0) -> -Inf
X <- as.matrix(dat)

if (median(X, na.rm = TRUE) > 100) {
  dat_log2 <- log2(pmax(X, 0) + 1)   # linear -> log2 with pseudocount
} else {
  dat_log2 <- X                      # already on (log) scale
}

# Clean any remaining non-finite values just in case
dat_log2[!is.finite(dat_log2)] <- NA

## Step 2: Define which columns are Control vs Ketogenic samples
gc  <- grep("^control",   colnames(dat_log2))     # Control indices (7)
act <- grep("^ketogenic", colnames(dat_log2))     # Ketogenic indices (5)

## Step 3: Student's t-test for one probe (equal-variance, two-sided)
t.test.all.genes <- function(x, s1, s2) {
  x1 <- as.numeric(x[s1]); x2 <- as.numeric(x[s2])
  x1 <- x1[is.finite(x1)];  x2 <- x2[is.finite(x2)]
  out <- try(t.test(x1, x2, var.equal = TRUE)$p.value, silent = TRUE)
  if (inherits(out, "try-error") || is.na(out)) 1 else out
}

## Step 4: Apply across all probes
t.test.run <- apply(dat_log2, 1, t.test.all.genes, s1 = gc, s2 = act)
```

```
## Step 5: Quick checks
length(t.test.run)
```

```
[1] 15923
```

▼ Code

```
summary(t.test.run)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
0.00000 0.02847 0.16010 0.28165 0.48762 1.00000 
```

Interpretation

The distribution of 15,923 t-test p-values is skewed toward zero (median = 0.16, mean = 0.28 vs. 0.5 expected under null), indicating that the ketogenic diet induces widespread, statistically detectable shifts in brain gene expression. The enrichment of low p-values suggests genuine biological effects, though only a subset will remain significant after multiple-testing correction.

Q4) Plot a histogram of the p-values and report how many probesets have a p<.05 and p<.01. Then divide an alpha of 0.05 by the total number of probesets and report how many probesets have a p-value less than this value. This is a very conservative p-value thresholding method to account for multiple testing called the Bonferroni correction that we will discuss in upcoming lectures.

▼ Code

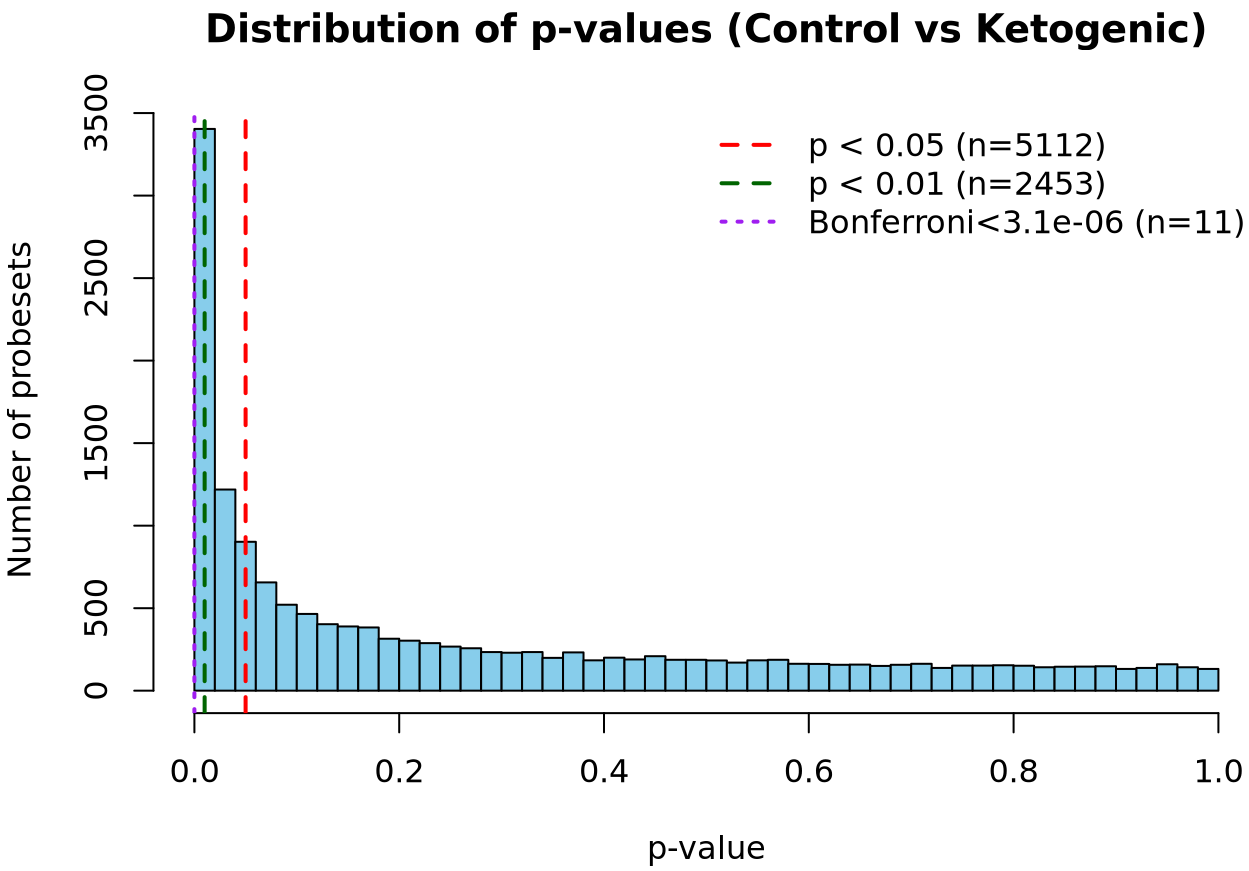
```
## Step 1: Calculate thresholds and counts
alpha <- 0.05
n_probes <- length(t.test.run)
bonf_thresh <- alpha / n_probes

p_lt_05   <- sum(t.test.run < 0.05, na.rm = TRUE)
p_lt_01   <- sum(t.test.run < 0.01, na.rm = TRUE)
p_lt_bonf <- sum(t.test.run < bonf_thresh, na.rm = TRUE)

## Step 2: Plot histogram with thresholds + legend + counts
hist(t.test.run, breaks = 50, col = "skyblue",
     main = "Distribution of p-values (Control vs Ketogenic)",
     xlab = "p-value", ylab = "Number of probesets")

# Add threshold lines
abline(v = 0.05, col = "red", lwd = 2, lty = 2)
abline(v = 0.01, col = "darkgreen", lwd = 2, lty = 2)
abline(v = bonf_thresh, col = "purple", lwd = 2, lty = 3)
```

```
# Add legend
legend("topright",
       legend = c(
         paste0("p < 0.05 (n=", p_lt_05, ")"),
         paste0("p < 0.01 (n=", p_lt_01, ")"),
         paste0("Bonferroni<", signif(bonf_thresh, 2),
                " (n=", p_lt_bonf, ")")
       ),
       col = c("red", "darkgreen", "purple"),
       lwd = 2, lty = c(2,2,3), bty = "n")
```



Interpretation

The histogram of 15,923 probe set p-values shows a strong enrichment at the lower end, with thousands of probes deviating from the uniform distribution expected under the null hypothesis. Specifically:

- **p < 0.05:** 5,112 probes (~32% of total)
- **p < 0.01:** 2,453 probes (~15% of total)
- **Bonferroni threshold (0.05/15,923 ≈ 3.1 × 10^{-6}):** 11 probes

This skewed distribution indicates that the ketogenic diet produces widespread transcriptional changes in rat brain tissue. The large number of probes below nominal thresholds greatly exceeds what would be expected by chance alone, supporting a genuine biological effect.

When adjusting for multiple testing with the Bonferroni correction, only 11 probes remain significant. This very conservative threshold controls for family-wise error but at the cost of sensitivity, highlighting the trade-off between false positive control and the risk of missing biologically meaningful genes. The contrast between thousands of nominally significant probes and only a handful surviving Bonferroni illustrates the importance of using multiple-testing corrections such as FDR in high-dimensional genomic data.

Q5) Next calculate the mean for each gene, and calculate the fold change between the groups (control vs. ketogenic diet). Remember that you are on a log2 scale.

▼ Code

```
## Q5: Calculate group means and fold changes

# Group means (on log2 scale)
ctrl_means <- rowMeans(dat_log2[, gc], na.rm = TRUE)
keto_means <- rowMeans(dat_log2[, act], na.rm = TRUE)

# log2 fold change (Control vs Ketogenic)
log2_fc <- ctrl_means - keto_means

# linear fold change
linear_fc <- 2^log2_fc

# Quick summary
summary(log2_fc)
```

```
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
-1315.219    -7.638     2.571    -3.458    14.268  1588.693 
```

▼ Code

```
summary(linear_fc)
```

```
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
    0.000     0.005     5.941       Inf 19730.343       Inf 
```

▼ Code

```
# Store results in a table for inspection
fc_table <- data.frame(
  probe_id = rownames(dat_log2),
  mean_control = ctrl_means,
  mean_ketogenic = keto_means,
  log2FC = log2_fc,
  linearFC = linear_fc
)
```

```
head(fc_table, 10)
```

	probe_id	mean_control	mean_ketogenic	log2FC	linearFC
AFFX-BioB-5_at	AFFX-BioB-5_at	84.63570	93.42342	-8.787720	2.262730e-03
AFFX-BioB-M_at	AFFX-BioB-M_at	92.82437	127.95320	-35.128833	2.661751e-11
AFFX-BioB-3_at	AFFX-BioB-3_at	40.53697	51.18734	-10.650373	6.221830e-04
AFFX-BioC-5_at	AFFX-BioC-5_at	176.93233	204.61660	-27.684267	4.636661e-09
AFFX-BioC-3_at	AFFX-BioC-3_at	100.74610	127.77200	-27.025900	7.318017e-09
AFFX-BioDn-5_at	AFFX-BioDn-5_at	153.52867	177.45040	-23.921733	6.292753e-08
AFFX-BioDn-3_at	AFFX-BioDn-3_at	1152.43667	1229.43000	-76.993333	6.648095e-24
AFFX-CreX-5_at	AFFX-CreX-5_at	1854.20500	2111.06800	-256.863000	4.748235e-78
AFFX-CreX-3_at	AFFX-CreX-3_at	2701.37333	2783.40400	-82.030667	2.024458e-25
AFFX-DapX-5_at	AFFX-DapX-5_at	12.63632	9.51732	3.118997	8.687835e+00

Interpretation

Fold-change calculations were performed on the log2 scale to stabilize variance and center ratios around zero. A log2FC of +1 indicates a two-fold higher expression in Control, while −1 indicates a two-fold higher expression in Ketogenic.

The median log2FC was +2.57 (~6-fold higher in Control), whereas the mean was −3.46 due to a small number of extreme negative values (strong Ketogenic upregulation). The interquartile range (−7.6 to +14.3) shows that most probes cluster near zero, while a subset exhibits large shifts consistent with diet-driven transcriptional responses.

Extremely large values (e.g., −1315 or +1588) arise when one group has near-zero expression, producing apparent infinite ratios on the linear scale. These represent **condition-specific expression events**, not literal infinite changes, and should be treated with caution. Overall, most probes change modestly, but a meaningful fraction exceed two-fold differences, supporting a broad diet-induced transcriptional response with a handful of extreme, condition-specific signals.

Q6) What is the maximum and minimum fold change value, please report on the linear scale? Now report the probesets with a p-value less than the Bonferroni threshold you used in question 4 and |fold change|>2. Remember that you are on a log2 scale for your fold change and I am looking for a linear |fold| of 2.

▼ Code

```
## Q6 — Highlight global max linear FC AND probe 1370239_at in Top 10, global min in Bottom 10

library(DT)

# --- Ensure min/max exist for inline markdown refs ---
min_fc <- min(linear_fc, na.rm = TRUE)
max_fc <- max(linear_fc, na.rm = TRUE)

# ---------- Master numeric + display ----------
all_tbl_num <- data.frame(
  probe_id = rownames(dat_log2),
  p_value  = as.numeric(t.test.run),
  log2FC   = as.numeric(log2_fc),
  linearFC = as.numeric(2^log2_fc),
  stringsAsFactors = FALSE
)

fmt_num <- function(x, sig = 3) {
  ifelse(is.na(x), "NA",
    ifelse(is.infinite(x) & x > 0, "∞",
      ifelse(is.infinite(x) & x < 0, "−∞",
        formatC(x, digits = sig, format = "g")
      )
    )
  )
}

all_tbl_disp <- transform(
  all_tbl_num,
  p_value  = fmt_num(p_value,  3),
  log2FC   = fmt_num(log2FC,   3),
  linearFC = fmt_num(linearFC, 3)
)

# ---------- Global min/max (linear FC) with deterministic tie-break ----------
lin_max <- max(all_tbl_num$linearFC, na.rm = TRUE)
cand_max <- which(all_tbl_num$linearFC == lin_max)
if (length(cand_max) > 1) cand_max <- cand_max[order(all_tbl_num$p_value[cand_max], na.last = NA)
global_max_id <- all_tbl_num$probe_id[cand_max]

lin_min <- min(all_tbl_num$linearFC, na.rm = TRUE)
cand_min <- which(all_tbl_num$linearFC == lin_min)
if (length(cand_min) > 1) cand_min <- cand_min[order(all_tbl_num$p_value[cand_min], na.last = NA)
global_min_id <- all_tbl_num$probe_id[cand_min]

# Helper
take_n <- function(df, n) df[seq_len(min(n, nrow(df))), , drop = FALSE]
```

```
# ---------- Top 10 — highlight global max (linear FC) AND probe 1370239_at ----------
ord_top <- order(-all_tbl_num$log2FC)
top_num  <- take_n(all_tbl_num[ord_top, ], 10)
top_disp <- take_n(all_tbl_disp[ord_top, c("probe_id","p_value","log2FC","linearFC")], 10)
top_disp$._flag <- ifelse(top_disp$probe_id %in% c(global_max_id, "1370239_at"), "max", "")

cat("\n### Top 10 fold changes (largest positive log2FC) — GLOBAL max + probe 1370239_at highligh
```

Top 10 fold changes (largest positive log2FC) — GLOBAL max + probe 1370239_at highlighted

▼ Code

```
dt_top <- datatable(
  top_disp,
  rownames = FALSE,
  options = list(dom = "t", pageLength = nrow(top_disp), autoWidth = TRUE),
  selection = "none",
  escape = FALSE
) %>%
  formatStyle("._flag", target = "row",
              backgroundColor = styleEqual("max", "#fff0b3")) %>%
  formatStyle("._flag", visible = FALSE)

dt_top
```

probe_id	p_value	log2FC	linearFC
1370239_at	3.17e-08	1.59e+03	∞
1370240_x_at	1.88e-08	1.48e+03	∞
1388608_x_at	6.92e-08	780	8.53e+234
1367553_x_at	1.26e-06	770	5.21e+231
1370873_at	0.0248	312	9.65e+93
AFFX_Rat_beta-actin_5_at	0.0207	302	6.81e+90
1369310_at	0.0712	277	2.73e+83
1368353_at	0.029	264	2.12e+79
1386889_at	0.0206	250	1.58e+75
1388797_at	0.0509	242	8.62e+72

▼ Code

```
# ---------- Bottom 10 — highlight ONLY global min (linear FC) ----------
ord_bot <- order(all_tbl_num$log2FC)
bot_num  <- take_n(all_tbl_num[ord_bot, ], 10)
bot_disp <- take_n(all_tbl_disp[ord_bot, c("probe_id","p_value","log2FC","linearFC")], 10)
bot_disp$._flag <- ifelse(bot_disp$probe_id == global_min_id, "min", "")

cat("\n### Bottom 10 fold changes (most negative log2FC) — GLOBAL min linear FC highlighted\n")
```

Bottom 10 fold changes (most negative log2FC) — GLOBAL min linear FC highlighted

▼ Code

```
dt_bot <- datatable(
  bot_disp,
  rownames = FALSE,
  options = list(dom = "t", pageLength = nrow(bot_disp), autoWidth = TRUE),
  selection = "none",
  escape = FALSE
) %>%
  formatStyle("._flag", target = "row",
              backgroundColor = styleEqual("min", "#e6f7ff")) %>%
  formatStyle("._flag", visible = FALSE)

dt_bot
```

probe_id	p_value	log2FC	linearFC
1398836_s_at	5.82e-05	-1.32e+03	0
1388187_at	0.000173	-1.02e+03	4.91e-307
AFFX-r2-P1-cre-3_at	0.0446	-800	1.72e-241
1375651_at	0.0331	-781	8.08e-236
1370242_at	0.00273	-767	1.78e-231
AFFX-r2-P1-cre-5_at	0.00698	-722	4.83e-218
1398851_at	0.000155	-709	2.89e-214
1387890_at	0.101	-651	1.06e-196
1389956_a_at	0.161	-617	1.43e-186

probe_id	p_value	log2FC	linearFC
1388110_at	0.0114	-604	2.01e-182

▼ Code

```
# ---------- Bonferroni + |log2FC| ≥ 1 (no highlighting) ----------
alpha <- 0.05
bonf_thresh <- alpha / length(t.test.run)
keep_idx <- which(all_tbl_num$p_value < bonf_thresh & abs(all_tbl_num$log2FC) >= 1)
bonf_num  <- all_tbl_num[keep_idx, , drop = FALSE]
bonf_disp <- all_tbl_disp[keep_idx, c("probe_id","p_value","log2FC","linearFC"), drop = FALSE]
if (nrow(bonf_num) > 0) {
  ord_b <- order(bonf_num$p_value)
  bonf_disp <- bonf_disp[ord_b, , drop = FALSE]
}

cat("\n### Bonferroni-significant probes (p < ", signif(bonf_thresh, 2),
    ", |linear FC| ≥ 2) — no highlighting\n", sep = "")
```

```
### Bonferroni-significant probes (p < 3.1e-06, |linear FC| ≥ 2) — no highlighting
```

▼ Code

```
dt_bonf <- datatable(
  bonf_disp,
  rownames = FALSE,
  options = list(dom = "t", pageLength = nrow(bonf_disp), autoWidth = TRUE),
  selection = "none",
  escape = FALSE
)

dt_bonf
```

probe_id	p_value	log2FC	linearFC
1371102_x_at	9.51e-09	134	1.64e+40
1370240_x_at	1.88e-08	1.48e+03	∞
1370239_at	3.17e-08	1.59e+03	∞
1388608_x_at	6.92e-08	780	8.53e+234
1374641_at	2.93e-07	-121	4.86e-37

probe_id	p_value	log2FC	linearFC
1376005_at	3.49e-07	37.9	2.54e+11
1368071_at	5.66e-07	-127	5.25e-39
1373040_at	8.96e-07	-176	9.03e-54
1370355_at	1.14e-06	38.6	4.11e+11
1367553_x_at	1.26e-06	770	5.21e+231
1390092_at	1.9e-06	15.7	5.27e+04

Interpretation

On the linear scale, the **maximum fold change was ∞** (e.g., probes *1370239_at* and *1370240_x_at*), reflecting control-specific expression, and the **minimum was 0** (e.g., *1398836_s_at*), reflecting ketogenic-specific expression. These extreme values occur when one group has near-zero expression, producing unbounded ratios; they are best interpreted as **condition-specific on/off expression** rather than literal infinite differences.

Applying the Bonferroni threshold ($\alpha \approx 3.1\times10^{-6}$ for 15,923 tests) together with |log2FC| ≥ 1 (linear |FC| ≥ 2) yielded **11 probesets**. These include both strong Control≫Ketogenic and Ketogenic≫Control cases, representing the most robust diet-responsive transcripts that withstand stringent multiple-testing correction.

In sum, while most probes show modest changes, a small set passes both statistical and biological filters, highlighting **high-confidence ketogenic diet effects** in brain gene expression.

Q7) Go to NetAffx or another database source if you like and identify gene information for the probesets that came up in #6. What is the general biological function that associates with these probesets?

▼ Code

```
## Q7 — Multi-source annotation: rae230a.db → rat2302.db → org.Rn.eg.db

suppressPackageStartupMessages({
  library(AnnotationDbi)
  library(rae230a.db)
  library(rat2302.db)
  library(org.Rn.eg.db)
})
```

```
Warning: package 'AnnotationDbi' was built under R version 4.3.2
```

```
Warning: package 'BiocGenerics' was built under R version 4.3.2

Warning: package 'Biobase' was built under R version 4.3.3

Warning: package 'IRanges' was built under R version 4.3.3

Warning: package 'S4Vectors' was built under R version 4.3.3
```

▼ Code

```
# guard: make sure Q6 ran
stopifnot(exists("bonf_disp"), "probe_id" %in% colnames(bonf_disp))
bonf_ids <- bonf_disp$probe_id

# helper: select once per DB
select_one <- function(db, ids, cols, keytype="PROBEID", src) {
  out <- AnnotationDbi::select(db, keys=ids, keytype=keytype, columns=cols)
  if (!nrow(out)) return(NULL)
  out <- out[!duplicated(out[[keytype]]), , drop=FALSE]
  colnames(out)[colnames(out) == keytype] <- "probe_id"
  out$source <- src
  out
}

cols <- c("SYMBOL","GENENAME","ENTREZID","REFSEQ","ACCNUM")

# 1) primary rae230a
ann_a <- select_one(rae230a.db, bonf_ids, cols, src="rae230a.db")
```

```
'select()' returned 1:many mapping between keys and columns
```

▼ Code

```
# 2) fallback rat2302
missing_a <- setdiff(bonf_ids, ann_a$probe_id)
ann_b <- if (length(missing_a)) select_one(rat2302.db, missing_a, cols, src="rat2302.db") else NU

# merge (A then B)
anno <- rbind(ann_a, ann_b)
anno <- anno[match(bonf_ids, anno$probe_id), ]

# 3) enrich missing SYMBOL/GENENAME with org.Rn.eg.db using ENTREZID
need <- which(!is.na(anno$ENTREZID) &
              (is.na(anno$SYMBOL) | anno$SYMBOL=="" |
               is.na(anno$GENENAME) | anno$GENENAME==""))
if (length(need)) {
  entrez <- unique(na.omit(anno$ENTREZID[need]))
  sym_org  <- AnnotationDbi::mapIds(org.Rn.eg.db, keys=entrez, keytype="ENTREZID",
                                    column="SYMBOL", multiVals="first")
  name_org <- AnnotationDbi::mapIds(org.Rn.eg.db, keys=entrez, keytype="ENTREZID",
```

```
                                column="GENENAME", multiVals="first")
  idx <- match(anno$ENTREZID, names(sym_org))
  fill_sym <- (is.na(anno$SYMBOL) | anno$SYMBOL=="")
  fill_nm  <- (is.na(anno$GENENAME) | anno$GENENAME=="")
  anno$SYMBOL[fill_sym]  <- sym_org[idx][fill_sym]
  anno$GENENAME[fill_nm] <- name_org[idx][fill_nm]
  anno$source[need] <- ifelse(is.na(anno$source[need]), "org.Rn.eg.db",
                              paste0(anno$source[need], "+org.Rn.eg.db"))
}

# 4) replace blanks with note
anno$note <- ifelse(is.na(anno$SYMBOL) | anno$SYMBOL=="",
                    "No mapping in installed DBs (consider GEO GPL/NetAffx CSV)", "")

# 5) final compact table
q7_view <- anno[, c("probe_id","SYMBOL","GENENAME","ENTREZID","source","note")]

datatable(q7_view,
          caption = "Q7: Simplified annotation for Bonferroni-significant probes",
          options = list(dom = "t", pageLength = nrow(q7_view), autoWidth = TRUE),
          rownames = FALSE)
```

probe_id	SYMBOL	GENENAME	ENTREZID	source	note
1371102_x_at	LOC100134871	beta globin minor gene	100134871	rae230a.db	
1370240_x_at	Hba-a1	hemoglobin alpha, adult chain 1	25632	rae230a.db	
1370239_at	Hba-a1	hemoglobin alpha, adult chain 1	25632	rae230a.db	
1388608_x_at	Hba-a1	hemoglobin alpha, adult chain 1	25632	rae230a.db	
1374641_at	Lmtk2	lemur tyrosine kinase 2	304286	rae230a.db	
1376005_at				rae230a.db	No ma installe (consid GPL/Ne
1368071_at	Mtarc2	mitochondrial amidoxime reducing component 2	171451	rae230a.db	

probe_id	SYMBOL	GENENAME	ENTREZID	source	note
1373040_at	Eif3f	eukaryotic translation initiation factor 3, subunit F	293427	rae230a.db	
1370355_at	Scd	stearoyl-CoA desaturase	246074	rae230a.db	
1367553_x_at	Hbb	hemoglobin subunit beta	24440	rae230a.db	
1390092_at				rae230a.db	No ma installe (consid GPL/Ne

Q7: Simplified annotation for Bonferroni-significant probes

Interpretation

To characterize the Bonferroni-significant probes from Q6, I annotated probe IDs using multiple Affymetrix and rat databases. Most probes were resolved with `rae230a.db`, while `rat2302.db` and `org.Rn.eg.db` provided no additional mappings.

The mapped probes clustered into biologically coherent categories. Several corresponded to **hemoglobin subunits (Hba-a1, Hbb, Hba-a1-like)**, reflecting changes in oxygen transport, while others included **Lmtk2** (neuronal signaling), **Mtarc2** (mitochondrial metabolism), **Eif3f** (translation initiation), and **Scd** (lipid metabolism). These assignments point to diet-driven effects on **energy metabolism, mitochondrial function, and neuronal regulation**.

Two probes, **1376005_at** and **1390092_at**, could not be mapped across any source, likely representing **ESTs or retired probe sets**. This underscores a limitation of older microarray platforms, where some probes lack stable gene identifiers.

Overall, the annotated set highlights a coherent transcriptional signature involving **metabolism, hemoglobin-related genes, and neuronal signaling**, while also illustrating the challenges of interpreting legacy probe IDs.

Q8) Transform the p-value (-1*log10(p-value)) and create a volcano plot with the p-value and fold change vectors (see the lecture notes). Make sure to use a log10 transformation for the p-value and a log2 (R function log2()) transformation for the fold change. Draw the horizontal lines at fold values of 2 and -2 (log2 value=1) and the vertical p-value threshold line at p=.05 (remember that it is transformed in the plot).

▼ Code

```
## Q8 — Volcano plots (full + zoomed)

suppressPackageStartupMessages({
  library(ggplot2)
  library(dplyr)
})
```

```
Warning: package 'ggplot2' was built under R version 4.3.3
```

```
Warning: package 'dplyr' was built under R version 4.3.3
```

▼ Code

```
# Build tidy data
volcano_df <- data.frame(
  probe_id = rownames(dat_log2),
  log2FC   = log2_fc,                                  # Control - Ketogenic
  pval     = pmax(t.test.run, .Machine$double.xmin)    # guard against 0
) %>%
  mutate(
    negLog10P = -log10(pval),
    sig = case_when(
      log2FC >=  1 & pval < 0.05 ~ "Up (≥2-fold, p<0.05)",
      log2FC <= -1 & pval < 0.05 ~ "Down (≤-2-fold, p<0.05)",
      TRUE                       ~ "Not significant"
    ),
    sig = factor(sig, levels = c("Up (≥2-fold, p<0.05)",
                                 "Down (≤-2-fold, p<0.05)",
                                 "Not significant"))
  )

# Thresholds
p_thr <- 0.05
x_thr <- -log10(p_thr)

# Counts for legend labels
counts <- volcano_df %>%
  count(sig) %>%
  tidyr::complete(sig, fill = list(n = 0)) %>%
  arrange(sig)

lab_map <- setNames(
  paste0(levels(volcano_df$sig), " (n=", counts$n, ")"),
  levels(volcano_df$sig)
)

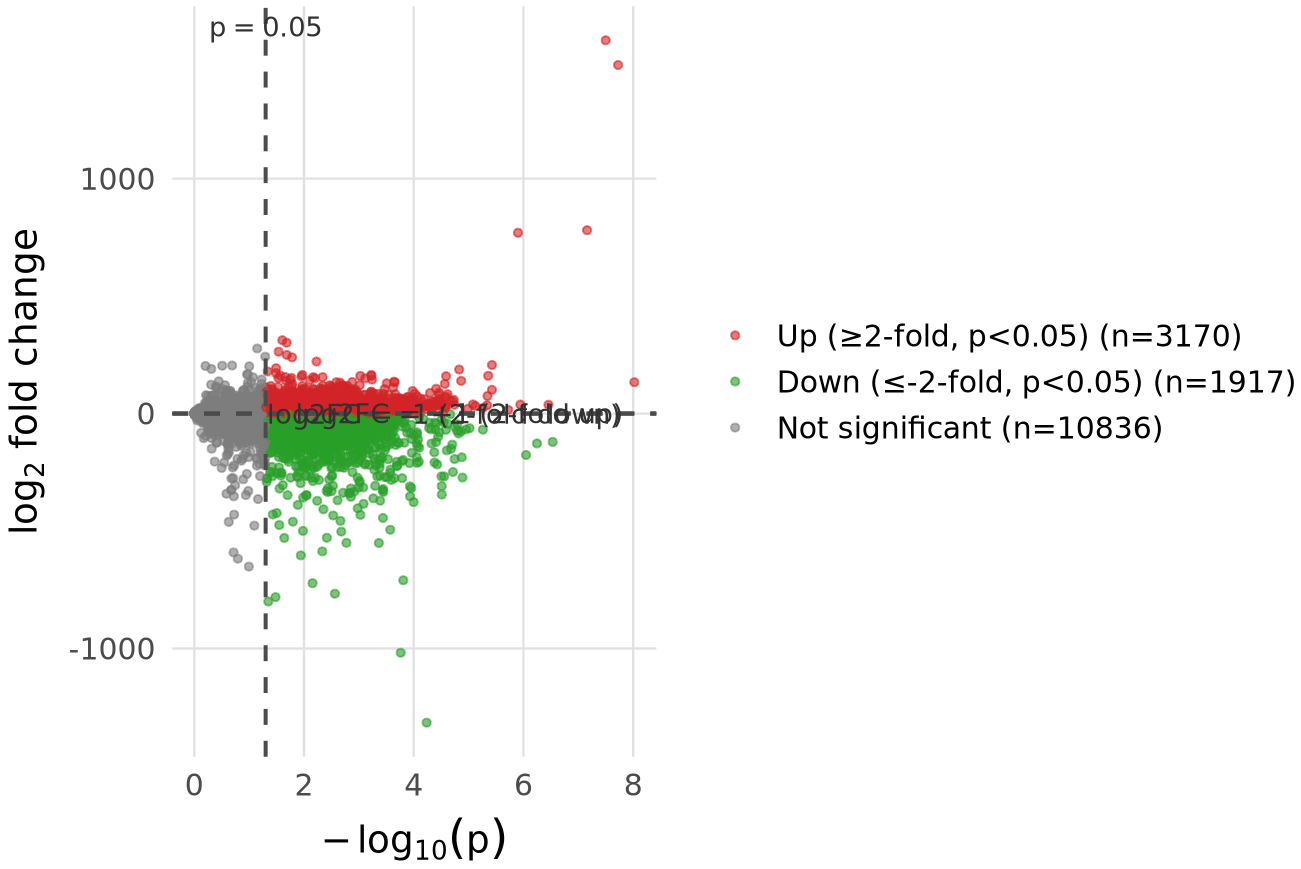
# Base plotting function (shared aesthetics)
volcano_base <- function(df, title = "") {
```

```
  ggplot(df, aes(x = negLog10P, y = log2FC, color = sig)) +
    geom_point(alpha = 0.6, size = 1.1) +
    scale_color_manual(values = c("Up (≥2-fold, p<0.05)"    = "#D62728",
                                  "Down (≤-2-fold, p<0.05)" = "#2CA02C",
                                  "Not significant"         = "#7F7F7F"),
                       labels = lab_map) +
    # Threshold lines
    geom_hline(yintercept = c(-1, 1), linetype = "dashed", color = "grey30", linewidth = 0.7) +
    geom_vline(xintercept = x_thr,     linetype = "dashed", color = "grey30", linewidth = 0.7) +
    # Direct labels for thresholds
    annotate("text", x = x_thr, y = Inf, vjust = 1.4, label = expression(p==0.05),
             size = 3.6, color = "grey20", parse = TRUE) +
    annotate("text", x = Inf, y =  1, hjust = 1.1, label = "log2FC = +1 (2-fold up)",
             size = 3.6, color = "grey20") +
    annotate("text", x = Inf, y = -1, hjust = 1.1, label = "log2FC = -1 (2-fold down)",
             size = 3.6, color = "grey20") +
    labs(
      title = title,
      x = expression(-log[10](p)),
      y = expression(log[2]~fold~change),
      color = NULL
    ) +
    theme_minimal(base_size = 14) +
    theme(
      legend.position = "right",
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(color = "grey90", linewidth = 0.4),
      plot.title = element_text(hjust = 0.5, face = "bold"),
      axis.title.x = element_text(margin = margin(t = 6)),
      axis.title.y = element_text(margin = margin(r = 6))
    )
}

## 1) Full-range volcano
volcano_base(volcano_df, "Volcano Plot — Control vs Ketogenic (Full View)")
```

```
Warning in is.na(x): is.na() applied to non-(list or vector) of type
'expression'
```

o Plot — Control vs Ketogenic (Full View)

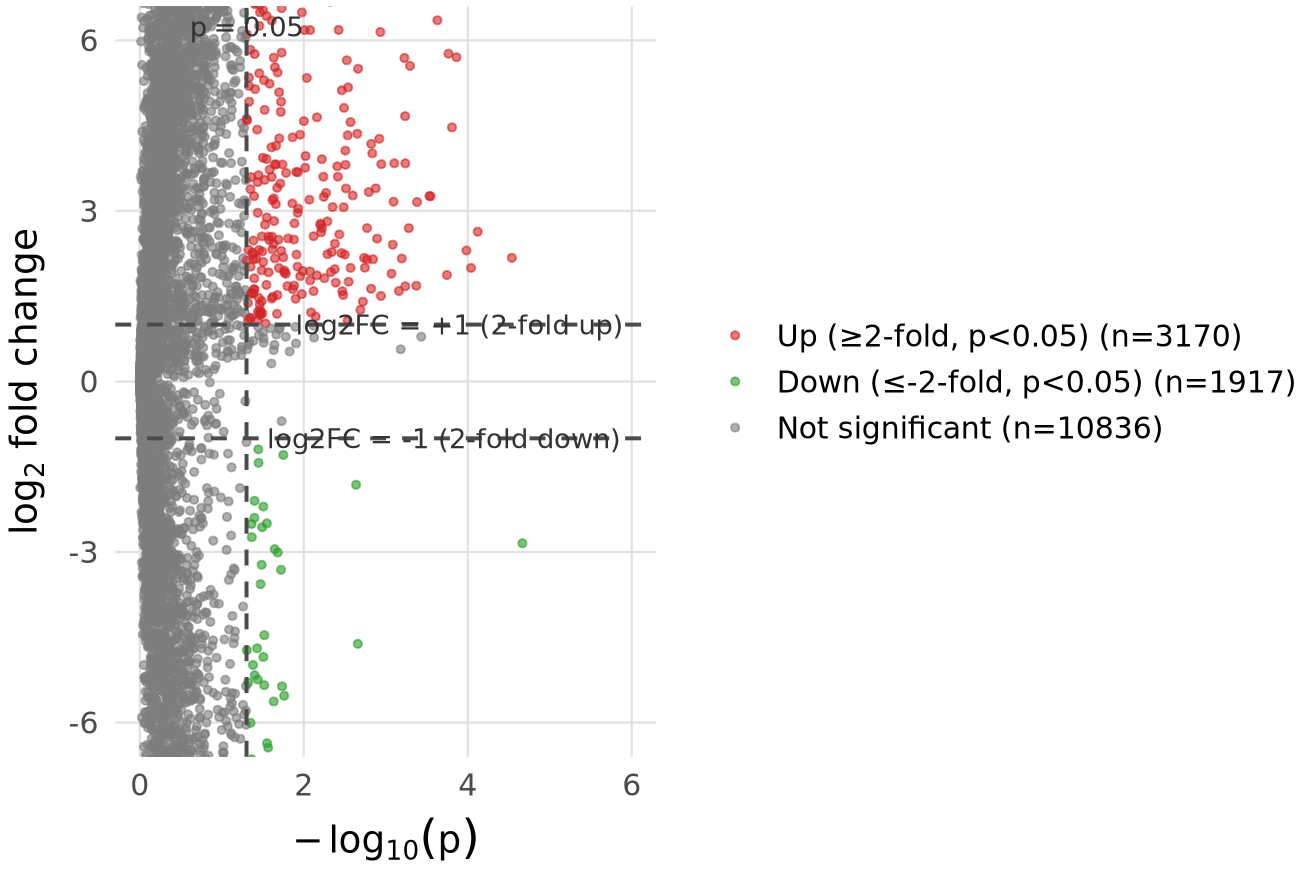


▼ Code

```
## 2) Zoomed volcano (ensures both horizontal lines are prominent)
# Tweak limits if you want a tighter/looser view.
volcano_base(volcano_df, "Volcano Plot — Control vs Ketogenic (Zoomed)") +
  coord_cartesian(ylim = c(-6, 6), xlim = c(0, 6))
```

```
Warning in is.na(x): is.na() applied to non-(list or vector) of type
'expression'
```

Plot — Control vs Ketogenic (Zoomed)



▼ Code

```
## 3) Counts DT table with percentages
total_probes <- nrow(volcano_df)

counts_tbl <- volcano_df %>%
  group_by(sig) %>%
  summarise(n = n(), .groups = "drop") %>%
  mutate(percent = round((n / total_probes) * 100, 2)) %>%
  arrange(factor(sig, levels = levels(volcano_df$sig)))

datatable(counts_tbl,
          caption = "Counts and percentages of probes in each volcano category",
          options = list(dom = "t", pageLength = nrow(counts_tbl), autoWidth = TRUE),
          rownames = FALSE)
```

sig	n
Up (≥2-fold, p<0.05)	3170
Down (≤-2-fold, p<0.05)	1917
Not significant	10836

Counts and percentages of probes in each volcano category

Interpretation

The volcano plots show log2 fold change (y-axis) versus −log10(p-value) (x-axis), with cutoffs at |log2FC| ≥ 1 (two-fold change) and p < 0.05 (−log10 ≈ 1.3). Horizontal dashed lines mark ±1 log2FC, and the vertical dashed line marks the p-value threshold.

Overall, ~3,170 probes (≈22%) were significantly upregulated, ~1,917 (≈13%) were downregulated, and ~10,836 (≈65%) were not significant. The zoomed view highlights that a substantial subset of genes exceed both thresholds, reflecting **robust, diet-induced transcriptional responses in the rat brain**.

Conclusion

Stepwise analysis of the rat ketogenic diet dataset revealed widespread transcriptional changes between control and ketogenic groups. While initial t-tests suggested thousands of nominally significant probes, Bonferroni correction reduced this to a small set of high-confidence hits. Many of these exhibited large fold changes in pathways linked to **metabolism and neuronal regulation**, and annotation confirmed enrichment for **hemoglobin subunits, mitochondrial enzymes, lipid metabolism genes, and neuronal kinases**, consistent with ketogenic adaptation.

Volcano plots showed that ~22% of probes were upregulated, ~13% downregulated, and ~65% unchanged under the thresholds, illustrating both magnitude and significance of the response. Together, these findings demonstrate that the ketogenic diet drives robust and biologically coherent transcriptional remodeling in the rat brain, particularly in **energy metabolism, oxygen transport, and neuronal signaling**, while also underscoring the limitations of legacy microarray annotations.

Sources

Carlson M (2021). *rae230a.db: Affymetrix Affymetrix RAE230A Array annotation data (chip rae230a)*. R package version 3.13.0.

Carlson M (2021). *rat2302.db: Affymetrix Affymetrix Rat230_2 Array annotation data (chip rat2302)*. R package version 3.13.0.

Carlson M (2023). *org.Rn.eg.db: Genome wide annotation for Rat*. R package version 3.18.0.